Architecture for Serious Games in Health Rehabilitation

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Abstract. Serious Games is a field of research that has been growing substantially with valuable contributions to many application areas. Traditional rehabilitation approaches are often considered repetitive and boring by the patients, resulting in difficulties to maintain their interest and to assure the completion of the treatment program. This paper describes a framework for the development of Serious Games that integrates a rich set of features including natural and multimodal interaction, social skills (collaboration and competitiveness) and progress monitoring which can be used to improve the designed games with direct benefits to the rehabilitation process. This improvement in the games' rehabilitation efficacy mainly arises from an increase in the patient's motivation when exercising the rehabilitation tasks due to the rich set of features provide by the framework.

Keywords: Serious Games, Rehabilitation, Architecture, Natural User Interfaces, Multimodal Interfaces, Health Informatics.

1 Introduction

Nowadays, the attention given to the development of tools for the rehabilitation of patients suffering from some kind of disability is growing. Additionally, the use of serious games as part of these rehabilitation tools is also increasing. Serious Games is a field of research that has been growing substantially and with valuable contributions in many areas.

As a multidisciplinary field of research, Serious Games can be applied into different problems in various and diverse areas such as: education, military, health rehabilitation. In such areas, computer games design and technology can be used to provide the user with an entertainment or enjoyment component, while attaining other main purpose that can be, respectively: educate, train (military strategies), and rehabilitate. The main purpose while playing the game is not the entertainment, or enjoyment, or fun, but another (more serious) purpose. In this research we focus the health rehabilitation domain.

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The costs in rehabilitation of a variety of disabilities resulting from diseases or traumatic incidents are high and contribute to the social costs growth. It has been shown that most of these patients sometimes get into depression as soon as the rehabilitation program begins, contributing to increase even more the associated healthcare costs. In previous studies [1-4] we discuss issues and identified features that could benefit the rehabilitation process using Serious Games. Based on that the main objective of this work is to describe our work regarding the design and implementation of an hardware/software architecture for the development of Serious Games in Health Rehabilitation, to augment patients' motivation during their recuperation plan and that integrates the set of features we find as relevant to this process.

The rest of the paper is structured as follows. Section 2 presents the motivation for using serious games in health rehabilitation. Following, section 3 presents the state of art in natural and multimodal user interfaces in rehabilitation, section 4 presents design considerations and system requirements for the developed architecture and section 5 describes the architecture. Section 6 presents results of user testing using the described architecture, current work and some directions for future work. Section 7 presents the major conclusions.

2 Serious Games for Health Rehabilitation

In the rehabilitation area, games can be used to increase the motivation of patients in the rehabilitation sessions, which is the main problem evidenced in traditional therapy sessions [5], caused by the repetitive nature of exercises.

Most studies on rehabilitation of patients with some disability or impairment show that an effective rehabilitation must be early, intensive and repetitive [5, 6]. As such, these rehabilitation approaches are often considered as repetitive and boring by the patients, resulting in difficulties in maintaining their interest and in assuring that they complete the treatment program [5]. On the other hand, due to its nature, games can motivate and engage the patients' attention and distract him from his rehabilitation condition, as they require some motor and cognitive activity, they have a story, and can offer feedback and levels of challenge and difficulty that can be adapted to the patient skills. However, designing a game with all the features that could benefit the outcome of patient's rehabilitation is a complex task. Computer games design can thus offer valuable contributions in the development of more effective games for rehabilitation programs. In that sense, the identification, classification and assessment of game features relevant in the health rehabilitation area is very important for designing the games.

In a previous study by Rego et al.[1], relevant criteria for classifying Serious Games in Health Rehabilitation were defined. The study shows that the adoption of Serious Games in rehabilitation area is very recent in general and the majority of reported researches include only prototypes and games in early stages of test and development. Additionally, their effectiveness is hard to validate due to the fact that few prototypes were tested with a significant number of real patients. Additionally, in the same study, interaction technology was pointed as an important factor that can influence game effectiveness in a rehabilitation program. Human Computer

Interaction (HCI) is a very important part of a systems design. The way the user communicates with the application/computer system determines its interactivity, accessibility and affects the user satisfaction with the system. In the rehabilitation area in particular, the interface can be used to improve interaction with the patient in a way that he can attain his goals more easily, more rapidly and with an higher level of satisfaction.

3 Natural and Multimodal User Interfaces in Rehabilitation

Recent directions in HCI design are toward the use of new methods of interaction: intelligent/adaptive, multimodal and natural. Natural interaction techniques aim to mimic real-world interaction by using user body movements, gestures, voice or sound input recognition and actions similar to those used in the physical world to accomplish the same task and to diminish the artificial communication devices for the interaction with the computer system. A multimodal interface integrates information from different input modalities (speech, gesture, writing, and others), providing a flexible use of input modes and an increase in the user degrees of freedom. In adaptive interfaces, the system tries to adapt itself to the way the user experiences the interaction. User experience can also be measured using several modalities of input, including biofeedback, information from voice, face and head. Brain signals contain information about user experience, along with other information measured from the body: bio signals as heart rate, body temperature, respiration, and muscle tension.

In the last years several prototypes using these new forms of interaction have been developed and published in rehabilitation area. Rego et al. [2] summarize relevant works of serious games for rehabilitation using natural forms of interaction.

A significant number of recent works involve motion tracking devices used to track patient's movements to detect and correct incorrect motion patterns during the rehabilitation exercises and/or to recover from several motor capabilities problems. The success of motion controlled game hardware and peripherals enables the application of games for rehabilitation, not only in hospitals and clinics, but also at home [7]. Several studies about the feasibility and results of using Wii and Wii sports game software showed similar improvements in functional outcomes, compared to conventional therapy, but the level of satisfaction and motivation reported by the users was higher using the Wii [8]. Some authors, besides developing custom applications to tailor the Wii game system with game tasks specific of rehabilitation, also use physical custom setups using the Wiimote as the base of the game system to obtain more affordable systems [9, 10], making it possible to be used at a home rehabilitation. Studies of balance and stability improvement on users with impairments typically involve the use of the Wii Balance Board integrated in custom applications [11]. On the other hand, actual mobile phone devices are available with integrated sensors enabling the development of applications that can sense motion and react to the mobile phone orientation. An example is given by Spina et al. [12] presenting an application that is based only in smartphone-integrated sensors to monitor and access the performance of the rehabilitation exercises in patients suffering from chronic pulmonary obstructive disease.

Kinect can be used in health and rehabilitation applications in several scenarios. Chang et al. [13] describe the use of Kinect in physical rehabilitation. The pilot study showed that the participants improve exercise performance during the intervention phases and that their motivation for physical rehabilitation has increased.

Data Gloves can be used as input devices to capture physical data such as bending of fingers, providing accurate data for motion capture that is interpreted by the software that accompanies the glove. In [14], the 5DT glove is connected to a PlayStation3 game console installed at their home and remotely monitored to study hand function in adolescents with hemiplegic cerebral palsy.

In a tangible user interface (TUI), the user interacts with digital information using physical objects of the real world as user interfaces, which is important in rehabilitation to diminish the cognitive load needed in the interaction. A relevant example is given in [15], describing the development of applications for motor rehabilitation based on a multi touch tabletop screen, providing tasks that closely mimic existing rehabilitation activities .

Touch/Multi-touch is an interaction modality that enables do detect contact of body parts (usually fingers) with a surface. The interface mechanism typically tracks the position of the contacts to recognize gestures. Additionally, the use of pressure information is present in some systems. In [16], a low-cost multi-touch surface platform was developed with several games for elderly cognitive training, and using a robust visual finger tracking algorithm.

Another emergent form of interaction is the one provided by BCI (brain-computer interfaces) systems enabling human-computer communication only by analysis of human brain signals. This technology can improve the daily life experience of people with neurological and motor control disorders after stroke or other traumatic brain disorders, by giving information of the current state of brain activity. On the other hand, it may increase the efficacy of a rehabilitation program and improve muscle control for the patient, since the use of brain signals can supplement an impaired muscle control. Its uses include helping disabled users to communicate with a machine [17], in neurofeedback games [18] and in video games as game controllers [19].

Facial expressions recognition systems and eye gaze tracking may also play a key role in HCI and natural user interfaces and in Serious Games in particular. The user facial expression gives a measure of his satisfaction or experience in the game and thus the level of interaction or experiences in the game can be adapted to that measure [20]. Eye gaze tracking is the process of detecting and tracking the point of where the user is looking at – the point of gaze. There are some examples of the use of eye gaze tracking in rehabilitation domain [21, 22].

Some of more recent projects are based on multimodal or adaptive interfaces, enabling various forms of inputs. Examples of these solutions can be seen in [23] and [24-32]. Faria et al. [24-32] uses the concept of an intelligent wheelchair that is controlled using high level commands in a multimodal interface that uses voice, facial expressions, head movements and joystick as the main input forms and users have the option to choose the input type that best fits their needs, enabling to increase their safety and comfort. Users can convey information also in the form of gestures. Trigueiros et al. [33-36] uses a gestures concept for a vision-based hand gesture recognition system that can be integrated in a HCI system.

4 Design Considerations and Requirements

In the current section we present our developed framework for serious games. We focus the presented discussion on the distinguishable features that we have considered to include on its design. In particular the developed architecture encompasses modules that allow for a set of natural and multimodal interaction, social skills (collaboration and competitiveness) and progress monitoring which can be used to improve the rehabilitation process.

4.1 Design of the Rehabilitation Tasks

The design of the rehabilitation tasks is a complex issue that depends on the kind of disabilities of the patient in rehabilitation. These disabilities can be diverse and so it would be necessary a program adapted to the patient needs in order to achieve the best results in the rehabilitation program. Based on that, the design of the rehabilitation tasks would require a multidisciplinary team from the medical field composed with doctors, nurses, psychologists, physiotherapists, and therapists. Also it would require professionals of the game design field. This is difficult since it implies communication between different professionals (and languages).

4.2 Interface with the System

Patients in rehabilitation have several cognitive and motor disabilities which make their interaction with the system more difficult. Eliminating or diminishing the use of artificial devices specific for interaction and enabling users to interact directly with objects of the real world (for example, using a racquet in a tennis game) without using an intermediary device, natural interfaces enable a more easy, intuitive, and realistic interaction. The use of only a single device can limit accessibility, which is why the combination of multiple input and output modalities is an important objective People in rehabilitation can have disabilities that prevent them from using traditional techniques, such as mouse or keyboard, and for these, alternative means of interaction should be considered. With a multimodal or adaptive interface the patient can choose the input form that suits better to his disability condition. Additionally, more immersive and realistic user experiences are provided.

4.3 Social Interaction

Games and game concepts have a stimulating effect on engagement in tasks. In addition, social interaction in games has demonstrated positive effects in healthy subjects in terms of player experience [37]. Social factors included in many games have had effects on people's enjoyment when playing the games. This may increase the social connectedness and motivation in rehabilitation patients to keep practicing the exercises they need. Social play can be achieved by the incorporation of competitive and collaborative tasks in the rehabilitation games [38].

4.4 Home Rehabilitation and Progress Monitoring

From literature review we can see work developments are being made towards the creation of applications that use low-cost devices to be accessible and affordable for people in general. In particular in the Rehabilitation domain this is an essential aspect.

The use of low cost systems can foster a home rehabilitation and create more opportunities for the patients to train the necessary exercises in their recuperation plan. This is another important aspect to have in consideration in our framework: the use of an interaction technology that can be inexpensive to be feasible for home use and thus promoting for the patients to play more frequently the games. The home rehabilitation tool should not be used to replace the therapist, but instead serve as a supplement to the assisted rehabilitation, allowing a recuperation plan done partly in home. The patient can then exercise the therapy plan without needing the continued assistance from a health professional, which may not be always available. Additionally, it can save the patient from having to move to the rehabilitation center so often, which in some cases might be problematic or uncomfortable, depending on factors such as the distance from the rehabilitation center and on the disability condition of the patient.

Facilities to detect, log, and analyze patient's tasks (cognitive or motor) prescribed by the therapists, in order to evaluate the patient's progress in the therapy are also key feature to include with respect to serious game for health rehabilitation.

5 Architecture for Serious Games for Health Rehabilitation

In order to demonstrate that Serious Games can be used to increase motivation of patients in rehabilitation, a framework was developed for Serious Games in Health Rehabilitation that integrates the set of features identified as relevant to improve the rehabilitation process, such as: natural and multimodal interaction, social skills (collaboration and competitiveness) and progress monitoring. Figure 1 presents the architecture diagram. The architecture comprises several distinct layers of input recognition. The first layer represents input modalities in raw form. The second layer represents an abstract recognition from the inputs received in raw form. In the third layer we can see a combination/fusion of more modalities among each other: gestures recognition, emotion recognition, representing higher levels of recognition processes.

The main modules of the architecture are:

Game Engine: is the game component that is most common to all games, representing the most generic component of the game logic.

Game Database: the most specific component of the games that represents the repository of all the games that are available for use in the rehabilitation session.

Social Networking: this module is responsible for creating the mechanisms for users (patients) to group together in social networks in order to communicate with each other and providing tools to mediate and facilitate social interactions.

Competition/Collaboration: responsible for creating the interaction mechanisms of competition and collaboration among users. It includes user modelling and profiling for establishing the handicaps with which each user will play with others, even if at different levels of the rehabilitation process.

User Management and Profiling: responsible for managing the information associated to the users, including their profiles, therapies they have to follow, state of the therapy, progress indicators, handicaps, etc.

Logging & Monitoring: module responsible for registering the users' logs, session duration, last difficulty level attained in each session in order to monitor the progress of each patient during the therapy.

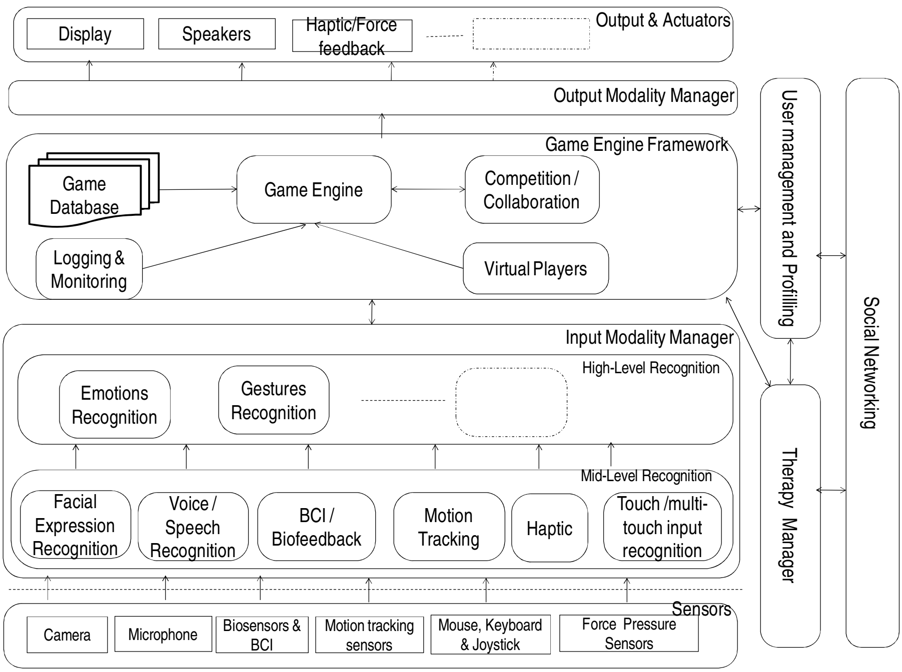

Fig. 1. Architecture Diagram

Virtual Players: module responsible for defining the agents that will assume the responsibility of users/players in the application sessions. This will enable some multi-player games even with only one real user.

Therapist Manager: provides a set of tools to configure, monitor and analyse the configuration of the prescribed therapy (the games that each patient will play, the duration of the sessions, the progress, etc.)

Input Modality Manager: manages the different input modalities and it can be adapted to the different disabilities of the users. It can be decomposed in two levels of recognition: in a mid-level we have individual forms of recognition: facial expressions, voice/speech, BCI/Biofeedback, motion tracking, Haptic and Tourch/multi-touch. In a higher level we can have a fusion of some of the modalities present in the mid-level that could compose a module of emotions recognition or gestures recognition, for example. The machine interpretation of human behaviour is very important in HCI, for achieving natural forms of interaction. In a mid-level recognition, users can communicate or interact with the machine by several forms that need to be interpreted by the machine. Users can convey messages in the form of body gestures, facial signals, speech, and emotions, among others. In a higher level of recognition, gestures recognition systems can have a multimodal nature, representing a fusion of the different input modalities present on the mid-level: gestures can be translated in: hand gestures; or body positions and movements that control a virtual environment on the screen, being captured by cameras or by the use of data sense gloves; or they can be a sequence of finger positions and movements in a multi-touch table; or they can be face gestures. Emotions have also a multimodal nature. They can be expressed through several modalities (and channels) of the mid-level recognition module: verbally by the use of emotional vocabulary or by expressing several non-verbal cues such as facial expressions, voice intonation, postures, gestures, and physiological changes. The decoding of these cues is essential to interpret the correct message. It is therefore necessary in a multimodal system to make a fusion of these different modalities to achieve a reliable (accurate) assessment.

Output Modality Manager: manages the different output modalities. It can include a display, sound device, haptic device and force-feedback equipment and it can be adapted to a specific game, environment or user.

The platform of the architecture provides virtual player and competition/collaboration modules that integrate with the social/users community component (social network).

6 Results and Discussion

We developed a prototype of the architecture to test the effect of the introduction of new forms of interaction in rehabilitation serious games. The prototype included three different interaction modalities: mouse, sound and motion. We tested the usability of the prototype with 20 healthy users and results showed that the sound and motion options were the modalities that provided the most involvement of the users and were the options they showed most interest to play again.

We are analyzing the multimodal capacity of the platform such that we can choose the input modality that best suits to the user capabilities or disability condition. One of the main contributions of our architecture is to allow that the user interface with the system is adaptive in the sense that it can be configured to the best input modality taking into consideration the user capabilities and the proposed tasks, reducing the physical and mental fatigue in the interaction with the system.

We also intend to study the efficacy of more natural modalities of interaction in the modules of competition/collaboration and virtual players. We point the originality of our architecture that derives from the fact that it provides a competitive/collaborative component, a multimodal integrated management component and virtual players that can surrogate real players when they do not exist. In that sense, in a multi-player game, the virtual player can substitute the patient when there are not patients available.

7 Conclusions

Several motion tracking devices have been introduced by the gaming industry, enabling the use of game consoles for home-based rehabilitation with some type of gesture based game play. Noticeable examples of the later are the Nintendo Wii with the Wiimote, PlayStation 2 EyeToy, PlayStation 3 Eye and Microsoft Kinect and PlayStation 3 Move systems. Modalities as speech recognition, facial expression recognition and touch input are also used in rehabilitation to become accessible by patients with several motor and cognitive impairments. Although many prototypes are being developed for rehabilitation serious games, the solutions available don´t fulfill yet all the requirements/features that must be defined for each type of therapy/patient. In order to demonstrate that Serious Games can be used to improve the rehabilitation process, a framework is described for the development of Serious Games that integrates a set of features identified as relevant to improve the rehabilitation process, such as: natural and multimodal interaction, social skills (collaboration and competitiveness) and progress monitoring.

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